Koneru Lakshmaiah Education Foundation (Deemed to be University)

Course Code: 22SDEC14R

Course Name: Embedded Prototype

(Skill Development Project-3)

A Project Report

On

SMART PADDLE WHEEL AERATOR SYSTEM WITH REAL TIME MONITORING FOR AQUACULTURE

SUBMITTED BY:

2300032670 – P.ANISH

2300040004 – B.SUMANTH

2300040006 - CH.CHANDU

2300040025 – S.HARINI

UNDER THE GUIDANCE OF

Dr A V Prabhu

Associate Professor-ECE

Green fields, Vaddeswaram – 522 502
Guntur Dist., AP, India.

K L E F

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

(DST-FIST Sponsored Department)

DECLARATION

The Project Report entitled **"SMART PADDLE WHEEL AERATOR SYSTEM WITH REAL TIME MONITORING FOR AQUACULTURE"** is a record of bonafide work of 2300032670 - P.ANISH , 2300040004 - B.SUMANTH , 2300040006 - CH.CHANDU , 2300040025 – S.HARINI ,submitted in partial fulfillment for the award of B.Tech in Electronics and Communication Engineering to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

K L E F

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

(DST-FIST Sponsored Department)

CERTIFICATE

This is to certify that the Project Report entitled "**SMART PADDLE WHEEL AERATOR SYSTEM WITH REAL TIME MONITORING FOR AQUACULTURE**" is being submitted by 2300032670 - P.ANISH , 2300040004 - B.SUMANTH , 2300040006 - CH.CHANDU , 2300040025 – S.HARINI in partial fulfilment for the award of B.Tech in Electronics and Communication Engineering to the K L University is a record of bonafide work carried out under our guidance and supervision.

The results embodied in this report have not been copied from any other departments/ University/Institute.

Signature of Supervisor

Signature of the HOD

Signature of the External Examiner

ACKNOWLEDGMENT

It is great pleasure for me to express my gratitude to our honourable President **Sri. Koneru Satyanarayana**, for giving the opportunity and platform with facilities in accomplishing the project report.

I express the sincere gratitude to our Vice Chancellor, **Dr G P S Varma** for his administration towards our academic growth.

I express the sincere gratitude to our Pro VCs, **Dr N Venkatram & Dr K Raja Sekara Rao** for their administration towards our academic growth.

I record it as my privilege to deeply thank our principal, **Dr.V. Rajesh** for providing us the efficient faculty and facilities to make our ideas into reality.

I express sincere gratitude to our pioneer **Dr I Govardhani, HOD-ECE** for his leadership and constant motivation provided in successful completion of our academic semester.

I express my sincere thanks to our project mentors **Dr A V Prabhu**, Associate Professor, ECE for their novel association of ideas, encouragement, appreciation, and intellectual zeal which motivated us to venture this project successfully.

Finally, it is pleased to acknowledge the indebtedness to all those who devoted themselves directly or indirectly to make this project report success.

ABSTRACT

The Smart Paddle Wheel Aerator System with Real-Time Monitoring for Aquaculture is designed to provide an efficient and intelligent solution for maintaining optimal water quality in fish ponds. It proposes an ESP32-based system integrated with turbidity, TDS, and pH sensors to continuously monitor essential water parameters. The collected data is displayed on an OLED screen and transmitted to the cloud for remote access and analysis. Based on real-time readings, the motor-driven paddle wheel aerator automatically regulates oxygen levels to maintain a healthy aquatic environment. By reducing manual intervention and optimizing energy usage, the system enhances productivity, ensures sustainable aquaculture practices and improves overall farm management efficiency.

In addition to improving dissolved oxygen control, the proposed system focuses on energy-efficient operation by running the aerator only when necessary, thereby reducing power consumption and operational costs. The design incorporates reliable hardware components and protective enclosures suitable for aquatic environments. Overall, the project provides a scalable, low-cost, and smart solution that supports precision aquaculture, minimizes losses due to poor water quality, and contributes to sustainable fish farming practices.

LITERATURE SURVEY

Aquaculture is one of the fastest-growing food production sectors worldwide, yet maintaining water quality remains a key challenge for sustainable fish farming. Several studies emphasize the importance of dissolved oxygen (DO), pH, turbidity, and total dissolved solids (TDS) as critical indicators of pond health, with poor water quality leading to increased fish stress and mortality (FAO, 2020). Traditional aeration systems, such as fixed paddle wheels or diffused air systems, are widely used but operate on manual control or fixed schedules, often leading to energy wastage and inconsistent oxygen regulation.

Other literature has focused on smart aerator designs — for instance, integrating paddle wheel aerators with automatic motor control to adjust speed based on DO levels. Researchers in the International Journal of Agricultural and Biological Engineering found that adaptive aeration not only improves oxygen distribution but also reduces electricity consumption compared to continuously running systems. Meanwhile, cloud-connected monitoring systems have been shown to enhance data collection and alert mechanisms, yet integration of multiple water quality parameters with automated paddle wheel control remains an area with limited practical demonstration.

In summary, while previous work establishes the effectiveness of IoT monitoring and the importance of maintaining pond water quality, gaps persist in seamlessly combining real-time multi-parameter sensing with automatic aerator actuation and cloud-based remote management**.** The Smart Paddle Wheel Aerator System addresses these limitations by offering a unified solution that intelligently monitors, controls, and manages aquaculture environments — enhancing productivity while reducing manual effort and energy use.

INDEX

S.NO	TITLE
1	Introduction
2	Aim of the Project
3	Proposed Methodology (Include Block Diagram/Flow Chart)
4	Components Explanation
5	Implementation Methodology (Circuit and Implementation)
6	Results and Discussions (With Screenshots)
7	Source Code
8	Conclusion and Future Scope

1.INTRODUCTION

Aquaculture plays a vital role in meeting the increasing global demand for fish and seafood products. However, maintaining proper water quality in fish ponds is one of the most critical challenges faced by farmers. Among various parameters, dissolved oxygen (DO) is the most essential factor influencing fish growth, metabolism, and survival. Low oxygen levels can cause stress, reduced productivity, and even large-scale fish mortality. Traditional aeration systems are generally operated manually or on fixed schedules, which may lead to inefficient oxygen control and excessive energy consumption.

To overcome these limitations, the Smart Paddle Wheel Aerator System with Real-Time Monitoring for Aquaculture is proposed as an automated and intelligent solution. This system continuously monitors key water quality parameters such as pH, turbidity, and TDS using sensors connected to an ESP32 microcontroller. Based on real-time data analysis, the system automatically controls a motor-driven paddle wheel aerator to maintain optimal pond conditions. Additionally, the integration of IoT technology enables remote monitoring through cloud platforms, allowing farmers to track water quality and system performance from anywhere. By combining automation, real-time monitoring, and energy-efficient operation, the project aims to improve aquaculture productivity, reduce manual intervention, and support sustainable fish farming practices.

2.AIM OF THE PROJECT

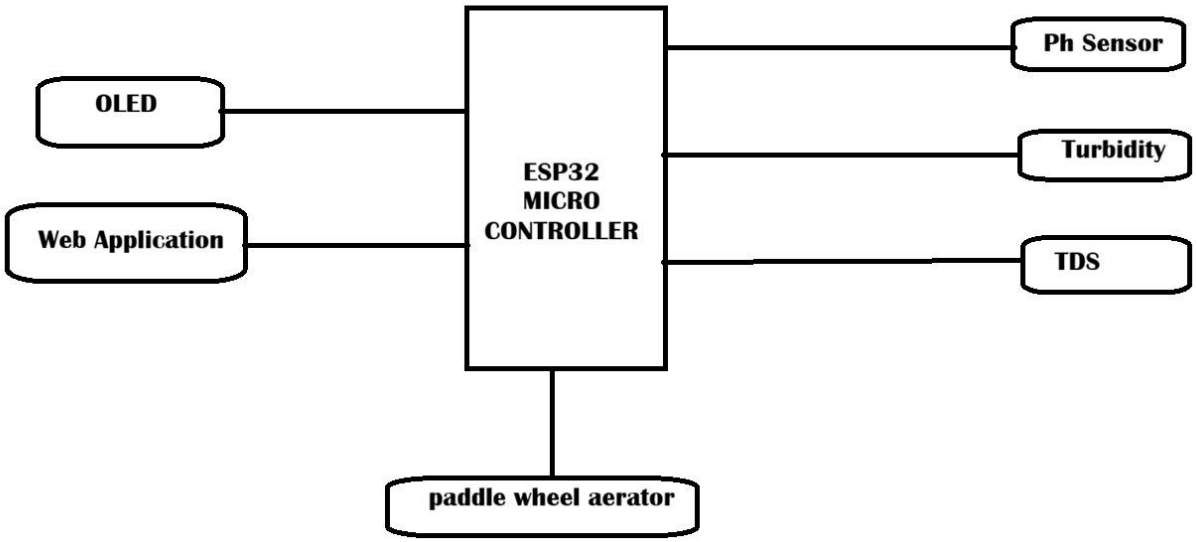
The aim of this project is to design and develop an intelligent Smart Paddle Wheel Aerator System capable of automatically maintaining optimal water quality conditions in aquaculture ponds. The system continuously monitors crucial parameters such as pH, turbidity, TDS, and oxygen-related conditions using sensors interfaced with an ESP32 microcontroller. Based on real-time data analysis, the controller automatically regulates the operation of the paddle wheel aerator to maintain sufficient dissolved oxygen levels, ensuring a healthy and stress-free environment for fish and shrimp growth.

Another major aim of the project is to integrate IoT-based real-time monitoring and cloud connectivity to enable remote access, data logging, and performance tracking. Farmers can monitor live pond conditions, analyze historical trends, and receive alerts during abnormal situations, allowing quick corrective actions. The system also focuses on improving energy efficiency by operating the aerator only when necessary, thereby reducing power consumption and operational costs. Additionally, the project aims to enhance scalability so that it can be implemented in small, medium, or large-scale fish farms. By combining automation, smart monitoring, and sustainable design, the system contributes to modernizing aquaculture practices, increasing productivity, minimizing fish mortality, and promoting long-term environmental sustainability.

3.METHODOLOGY

The Smart Paddle Wheel Aerator System with Real-Time Monitoring for Aquaculture is designed to maintain optimal dissolved oxygen levels in fish ponds. Water quality parameters such as dissolved oxygen, temperature, and pH are continuously monitored using sensors connected to a microcontroller. When the oxygen level drops below a preset threshold, the controller automatically activates the paddle wheel aerator to increase oxygen concentration in the water. The system also transmits real-time data to a cloud platform for remote monitoring through a mobile or web application. This automation improves energy efficiency, reduces fish mortality, and ensures better aquaculture management.

BLOCK DIAGRAM



4.COMPONENTS EXPLANATION

ESP32 Board

ESP32 is a Wi-Fi enabled microcontroller used as the main controller of the system.

It reads sensor data, controls the DC motor, and sends real-time information to the cloud.

DC Motor

The DC motor converts electrical energy into mechanical rotation.

It drives the paddle wheels to aerate the pond water when activated by the ESP32.

Paddle Wheels

Paddle wheels are rotating blades placed on the water surface.

They circulate water and increase oxygen transfer by agitating the pond water.

Turbidity Sensor

The turbidity sensor measures the clarity or cloudiness of water.

It helps monitor water quality by detecting suspended particles in the pond.

TDS Sensor

The TDS sensor measures the amount of dissolved salts and minerals in water.

It helps determine whether the water is suitable for healthy fish growth.

pH Sensor

The pH sensor measures the acidity or alkalinity of the pond water.

It ensures the water remains within the safe pH range for aquatic life.

OLED Display

The OLED display shows real-time sensor readings and system status.

It allows local monitoring of water parameters and motor operation.

Cloud

The cloud platform stores and displays sensor data remotely.

It enables farmers to monitor pond conditions anytime through the internet.

5.IMPLEMENTATION METHODOLOGY

The implementation of the Smart Paddle Wheel Aerator System with Real-Time Monitoring for Aquaculture is carried out in a systematic manner, involving hardware integration, software development, and system testing.

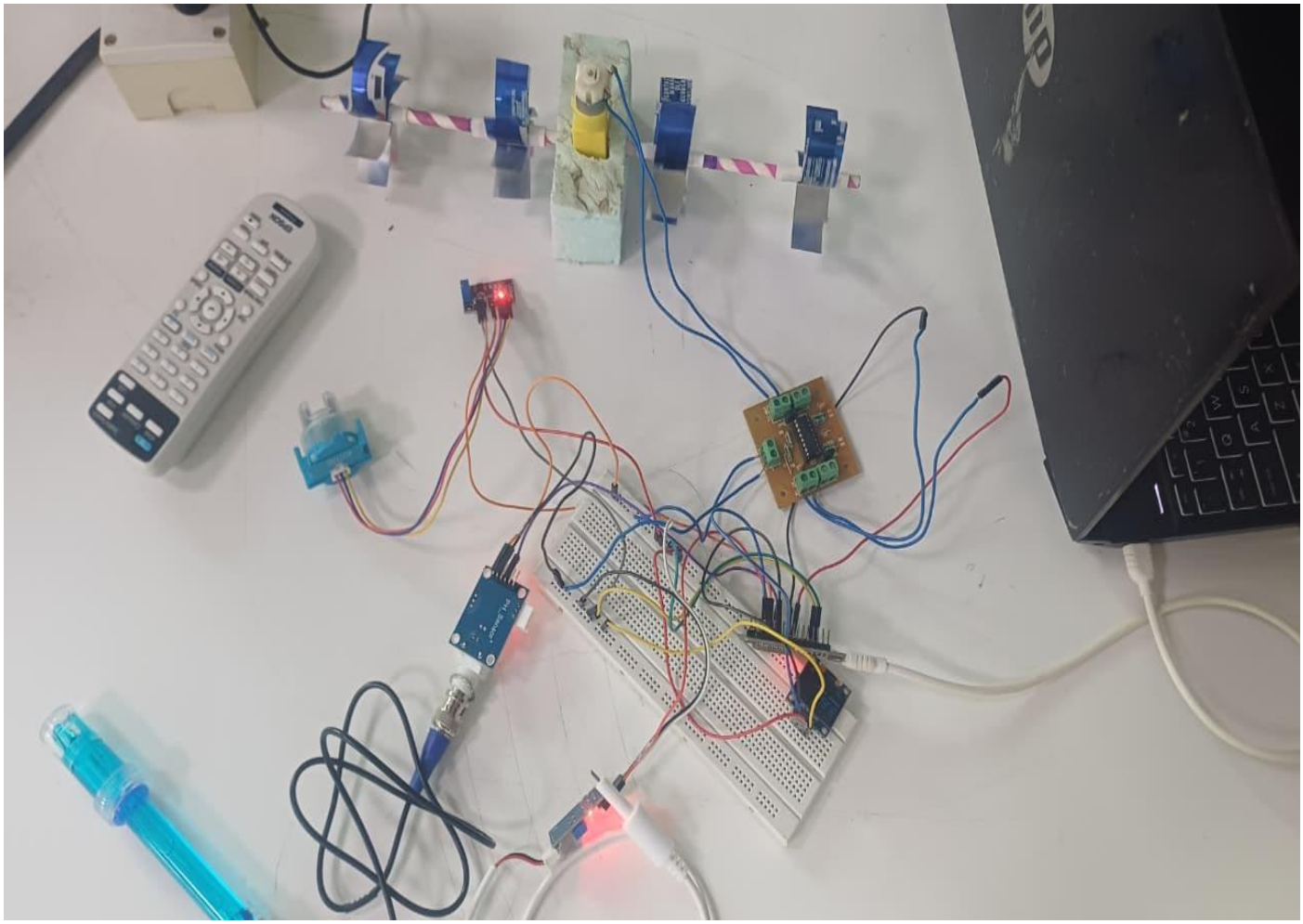
Initially, all hardware components such as the ESP32 board, turbidity sensor, TDS sensor, pH sensor, OLED display, DC motor, and paddle wheels are selected and assembled. The sensors are connected to the analog/digital input pins of the ESP32 to continuously measure water quality parameters. The DC motor is interfaced using a motor driver module to safely control the paddle wheel operation. Proper power supply arrangements and waterproof enclosures are provided to ensure safe and reliable operation in pond environments.

Next, embedded programming is developed using Arduino IDE for the ESP32. The program is designed to read sensor values, process the data, and compare the readings with predefined threshold levels. If the water parameters indicate poor quality or low oxygen-related conditions, the controller automatically activates the DC motor to drive the paddle wheels. The real-time data is displayed on the OLED screen for local monitoring.

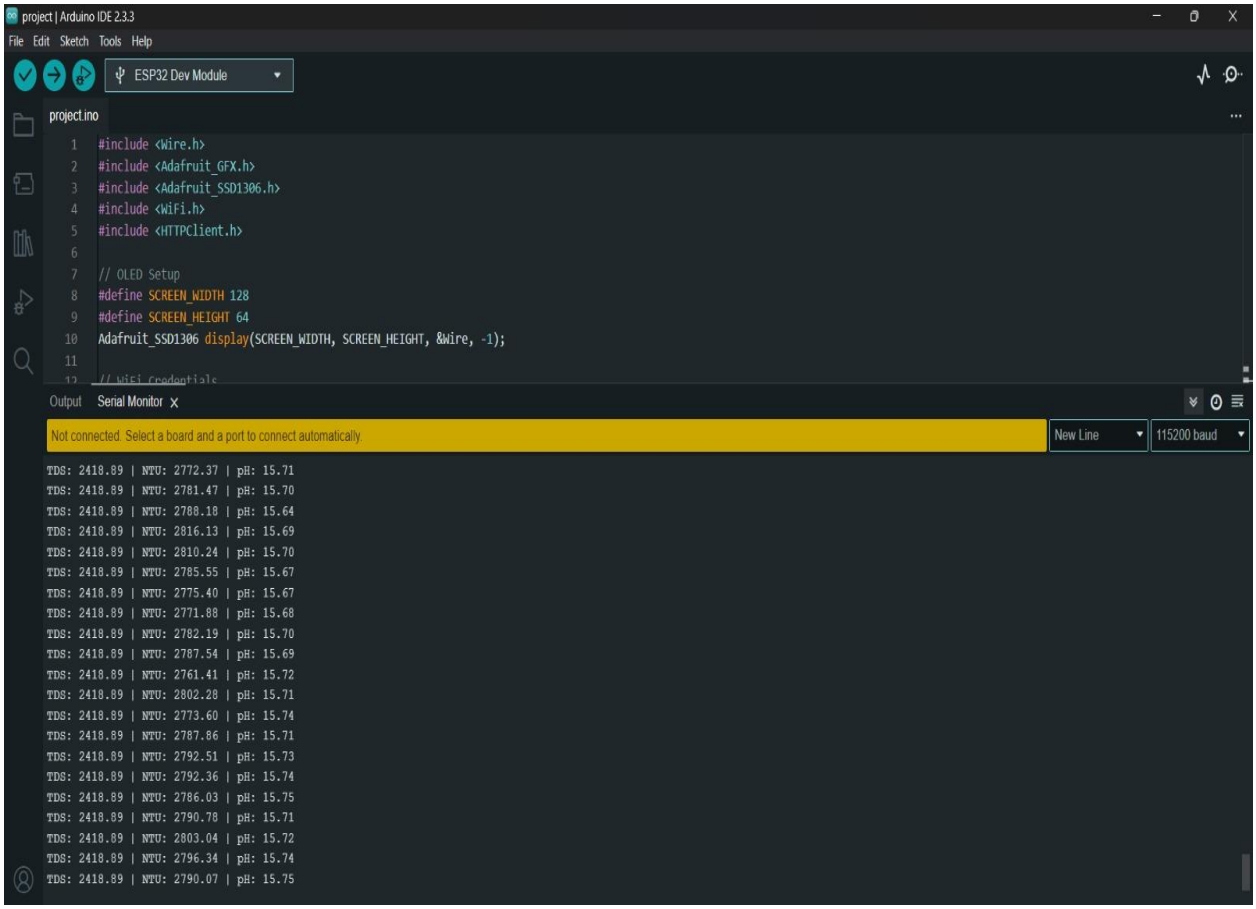
Finally, IoT integration is implemented by connecting the ESP32 to a Wi-Fi network. Sensor data is transmitted to a cloud platform for remote monitoring,

data logging, and analysis. The complete system is tested under different water conditions to evaluate performance, response time, accuracy, and energy efficiency. Through this structured implementation process, the system ensures effective automation and intelligent aquaculture management.

CIRCUIT DIAGRAM



6.RESULTS AND DISCUSSION



7.SOURCE CODE

```
#include <Wire.h>

#include <Adafruit_GFX.h>

#include <Adafruit_SSD1306.h>

#include <WiFi.h>

#include <HTTPClient.h>


// OLED Setup

#define SCREEN_WIDTH 128

#define SCREEN_HEIGHT 64

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);


// WiFi Credentials

const char* ssid = "project";

const char* password = "12345678";


// ThingSpeak

String apiKey = "9DEA80M3G1Y4FA46";

const char* server = "http://api.thingspeak.com/update";


// Sensor Pins

#define TDS_PIN   34

#define TURB_PIN  35

#define PH_PIN    32
```

```
// Motor Pins
#define MOTOR_IN1 26
#define MOTOR_IN2 27
#define MOTOR_ENA 14

// Thresholds
const float TDS_THRESHOLD  = 3000.0;
const float NTU_THRESHOLD  = 3000.0;
const float PH_LOW_LIMIT   = 0.0;
const float PH_HIGH_LIMIT  = 15.0;

// PWM
const int freq = 5000;
const int resolution = 8;

// Averaging
const int samples = 10;

void setup() {
  Serial.begin(115200);

  // Keep pulldown so pin becomes 0V if sensor removed
  pinMode(TDS_PIN, INPUT_PULLDOWN);
  pinMode(TURB_PIN, INPUT_PULLDOWN);
  pinMode(PH_PIN, INPUT_PULLDOWN);
```

```
  pinMode(MOTOR_IN1, OUTPUT);
  pinMode(MOTOR_IN2, OUTPUT);

  ledcAttach(MOTOR_ENA, freq, resolution);

  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println("OLED failed");
    while (1);
  }

  display.clearDisplay();
  display.display();

  WiFi.begin(ssid, password);
  Serial.print("Connecting to WiFi");

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\nWiFi Connected!");
}

// Averaging Function
float readAverage(int pin) {
```

```
  long total = 0;
  for (int i = 0; i < samples; i++) {
    total += analogRead(pin);
    delay(5);
  }
  return (float)total / samples;
}

void loop() {

  float avgTds = readAverage(TDS_PIN);
  float vTds = (avgTds / 4095.0) * 3.3;

  float avgTurb = readAverage(TURB_PIN);
  float vTurb = (avgTurb / 4095.0) * 3.3;

  float avgPh = readAverage(PH_PIN);
  float vPh = (avgPh / 4095.0) * 3.3;

  // ✅ Correct Sensor Disconnect Detection
  bool tdsMissing  = (vTds < 0.05);
  bool turbMissing = (vTurb < 0.05);
  bool phMissing   = (vPh < 0.05);

  bool sensorError = tdsMissing || turbMissing || phMissing;
```

```
if (sensorError) {
  runMotor(0);

  display.clearDisplay();
  display.setTextColor(WHITE);
  display.setTextSize(2);
  display.setCursor(10, 20);
  display.print("SENSOR");
  display.setCursor(10, 40);
  display.print("ERROR");
  display.display();

  Serial.println("Sensor Disconnected!");
  delay(200);
  return;
}

float tdsValue = (133.42 * pow(vTds, 3)
                - 255.86 * pow(vTds, 2)
                + 857.39 * vTds) * 0.5;

float ntu = -1120.4 * sq(vTurb)
          + 5742.3 * vTurb
          - 4353.8;

if (ntu < 0) ntu = 0;
```

```
  float phValue = 7 + ((2.5 - vPh) / 0.18);

  bool tdsDirty  = (tdsValue > TDS_THRESHOLD);
  bool turbDirty = (ntu > NTU_THRESHOLD);
  bool phDirty = (phValue < PH_LOW_LIMIT || phValue > PH_HIGH_LIMIT);

  bool waterDirty = tdsDirty || turbDirty || phDirty;

  if (waterDirty) {
    runMotor(200);
  } else {
    runMotor(0);
  }

  display.clearDisplay();
  display.setTextColor(WHITE);
  display.setTextSize(1);

  display.setCursor(0, 0);
  display.print("Water Quality Monitor");

  display.drawLine(0, 10, 128, 10, WHITE);

  display.setCursor(0, 15);
  display.print("TDS: ");
```

```
display.print((int)tdsValue);

display.setCursor(0, 25);
display.print("NTU: ");
display.print((int)ntu);

display.setCursor(0, 35);
display.print("pH: ");
display.print(phValue, 2);

display.setCursor(0, 50);
display.print("STATUS: ");
display.print(waterDirty ? "DIRTY" : "CLEAN");

display.display();

Serial.print("TDS: "); Serial.print(tdsValue);
Serial.print(" | NTU: "); Serial.print(ntu);
Serial.print(" | pH: "); Serial.println(phValue);

if (WiFi.status() == WL_CONNECTED) {
  HTTPClient http;

  String url = String(server) + "?api_key=" + apiKey +
               "&field1=" + String(tdsValue) +
               "&field2=" + String(ntu) +
```

```
                "&field3=" + String(phValue);

    http.begin(url);
    http.GET();
    http.end();
  }

  delay(15000);
}

void runMotor(int speed) {

  if (speed > 0) {
    digitalWrite(MOTOR_IN1, HIGH);
    digitalWrite(MOTOR_IN2, LOW);
  } else {
    digitalWrite(MOTOR_IN1, LOW);
    digitalWrite(MOTOR_IN2, LOW);
  }

  ledcWrite(MOTOR_ENA, speed);
}
```

8.CONCLUSION

The Smart Paddle Wheel Aerator System with Real-Time Monitoring for Aquaculture provides an efficient and intelligent solution for maintaining optimal pond water quality. By integrating sensors, an ESP32 microcontroller, and IoT technology, the system continuously monitors key parameters and automatically controls the aerator based on real-time conditions. This automation reduces manual effort, prevents sudden water quality deterioration, and ensures a healthy aquatic environment for fish growth.

The system also improves energy efficiency by operating the aerator only when required, thereby lowering operational costs. With real-time cloud monitoring and data logging, farmers can make informed decisions and respond quickly to unfavorable conditions. Overall, the project demonstrates a cost-effective, reliable, and scalable approach to modern aquaculture management.

FUTURE SCOPE

In the future, the system can be enhanced by integrating dissolved oxygen (DO) sensors for more precise oxygen control. Solar power with battery backup can be added to make the system energy-independent and suitable for remote areas.

Advanced features such as AI-based predictive analysis, multi-pond monitoring, automated feeding systems, and GSM-based alert notifications can further improve efficiency. With continuous improvements, the system can be expanded into a complete smart aquaculture management platform supporting large-scale and commercial fish farming operations.